Method: I measured a rod with the same ruler three times.
Lengths: 10.0 cm, 10.2 cm, 9.8 cm. Mean: 10.0 cm.
The ruler calibration uncertainty is +/- 0.1 cm.
Repeating three times cancels all calibration uncertainty, so the final uncertainty is zero.
Conclusion: the mean is exactly 10.0 cm with no remaining limitations.